

Conducting Forensic Investigations on Network Infrastructure (4e)

Digital Forensics, Investigation, and Response, Fourth Edition - Lab 09

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Time on Task:

6 hours, 48 minutes

Progress:

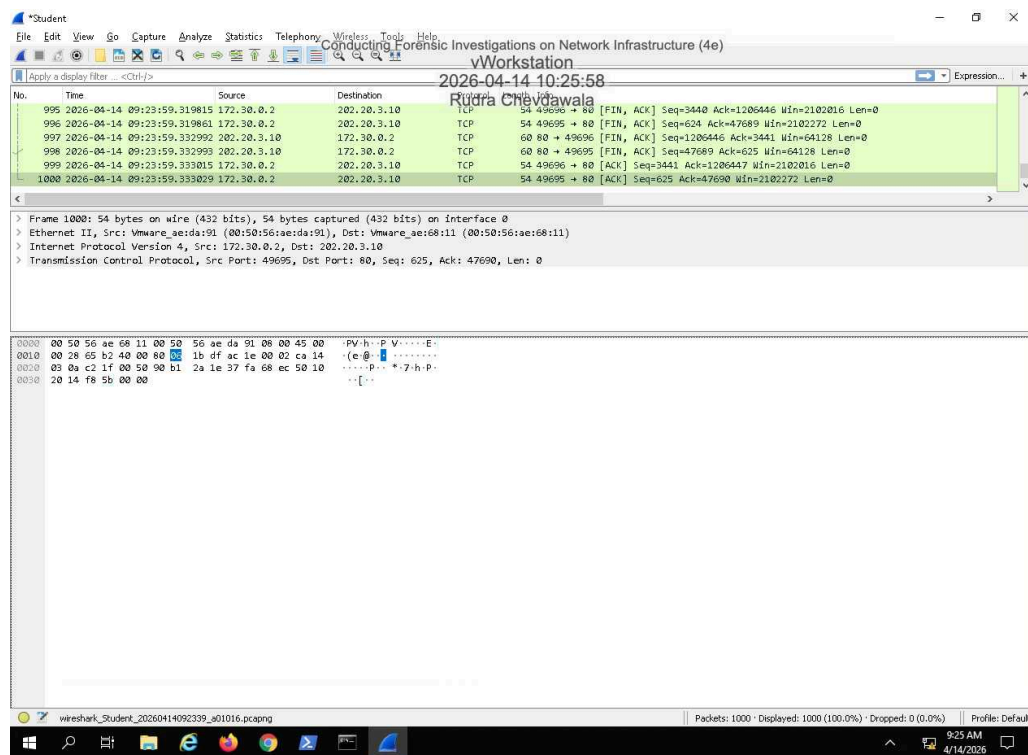
100%

Report Generated: Thursday, April 16, 2026 at 1:28 AM

Section 1: Hands-On Demonstration

Part 1: Perform Packet Capture and Analysis

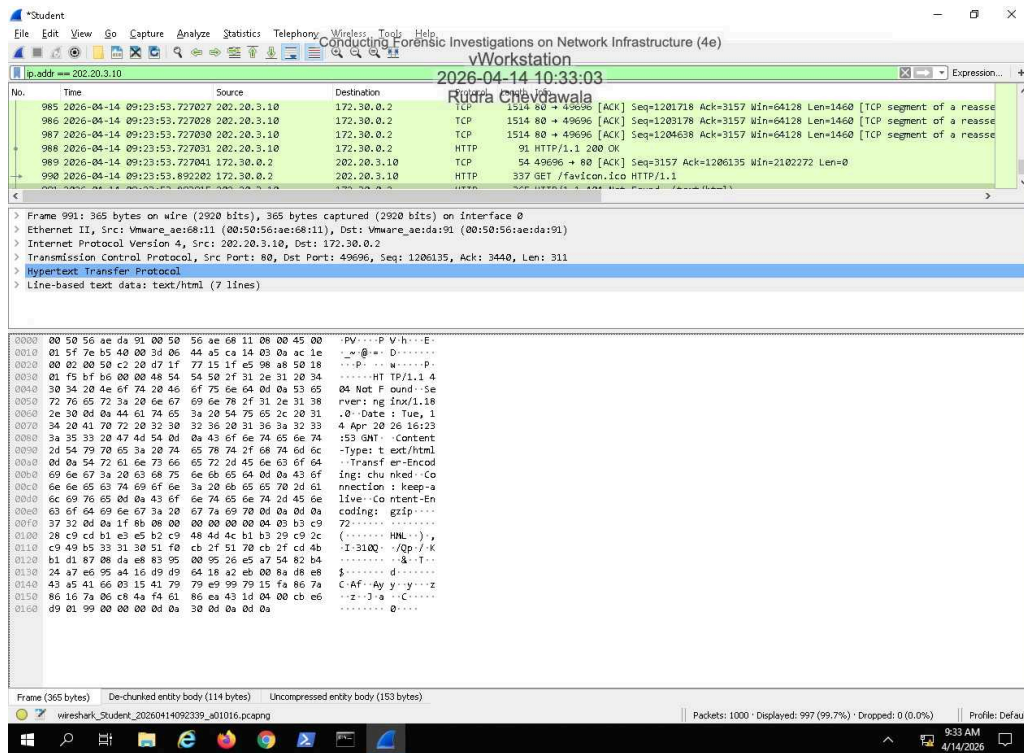
11. Make a screen capture showing the timestamp-sorted traffic.



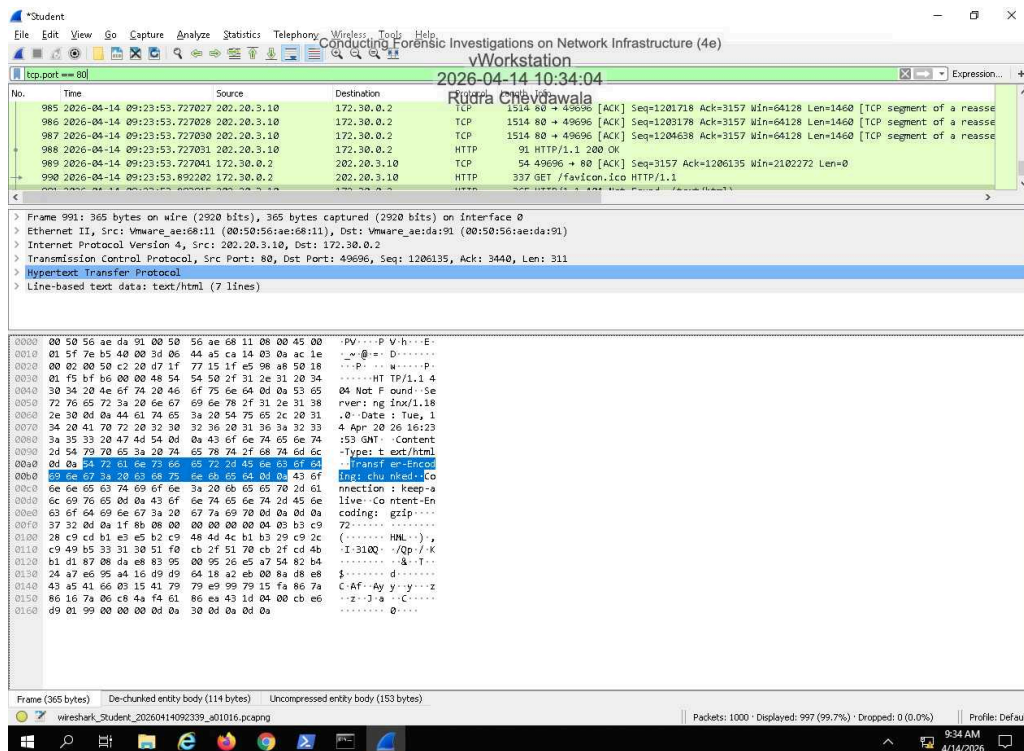
Conducting Forensic Investigations on Network Infrastructure (4e)

Digital Forensics, Investigation, and Response, Fourth Edition - Lab 09

13. Make a screen capture showing the IP-filtered traffic.



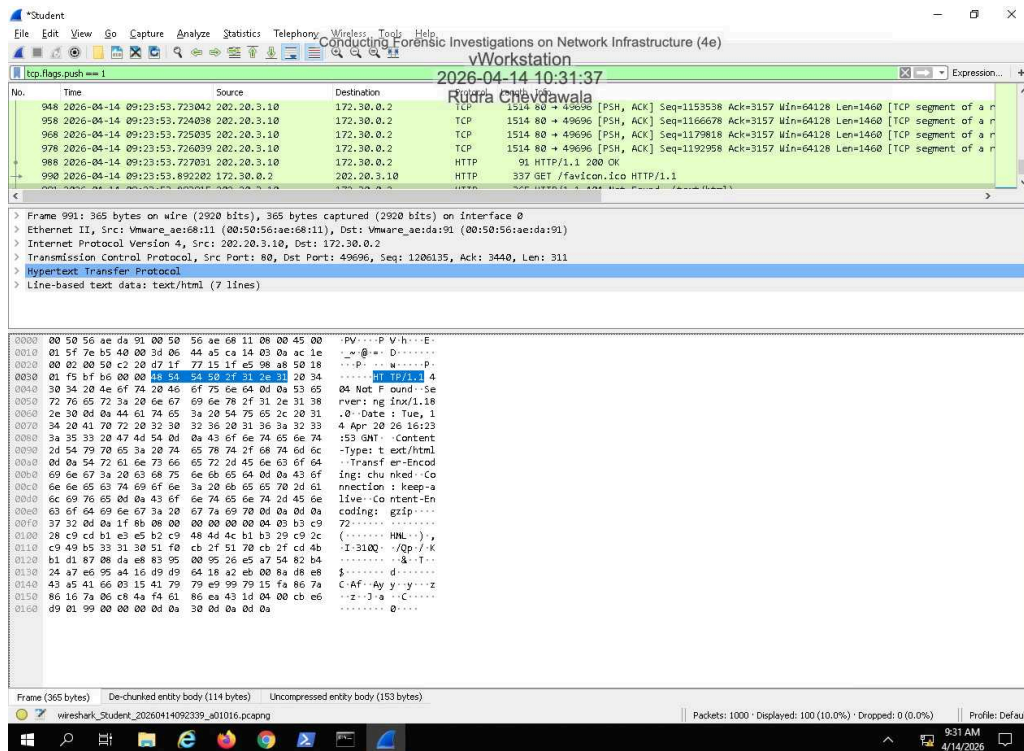
15. Make a screen capture showing the port-filtered traffic.



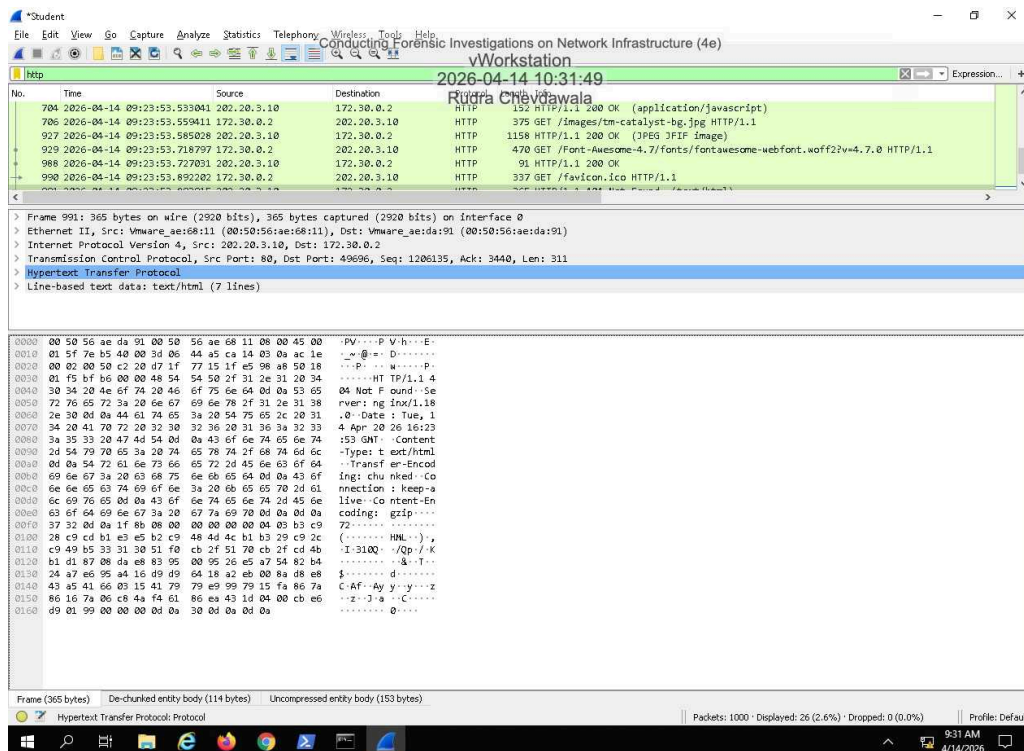
Conducting Forensic Investigations on Network Infrastructure (4e)

Digital Forensics, Investigation, and Response, Fourth Edition - Lab 09

17. Make a screen capture showing the TCP push flag-filtered traffic.

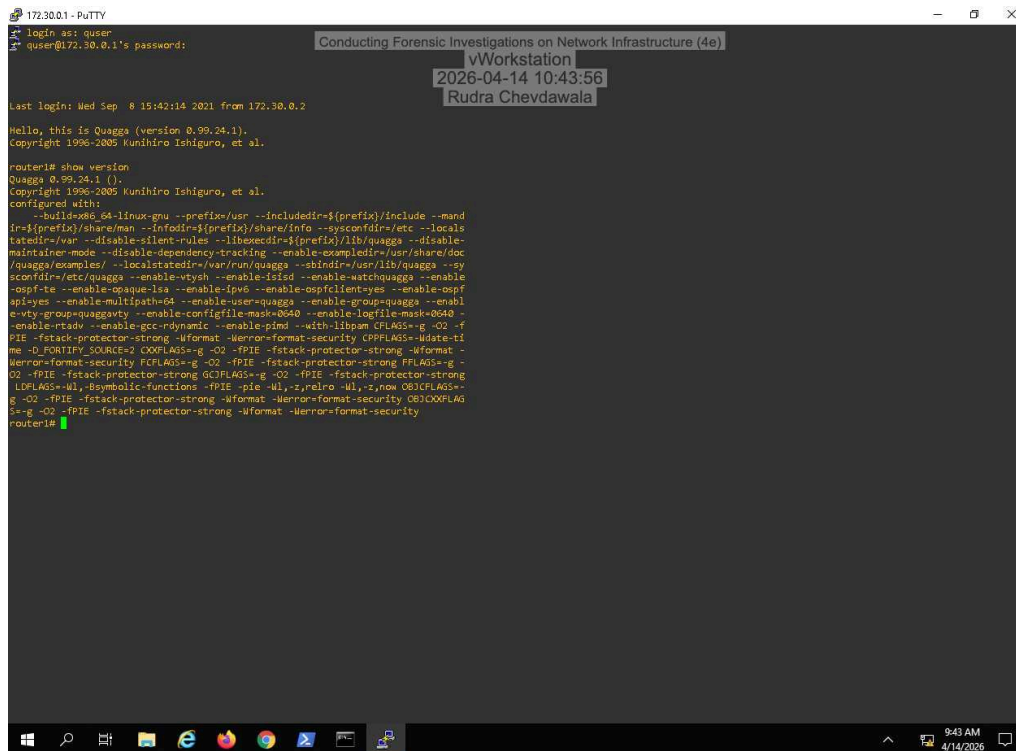


19. Make a screen capture showing the http-filtered traffic.



Part 2: Analyze a Router for Forensic Evidence

5. Make a screen capture showing the router's version output.



The screenshot shows a PuTTY terminal window titled '172.30.0.1 - PuTTY'. The user 'quser' has logged in and entered the password. The terminal displays the output of the 'show version' command on a Quagga router. The output includes the router's name 'router1a', the Quagga version '0.99.24.1', and a detailed list of configuration options used during the build process, such as '--enable-ipv6', '--enable-ospf', and '--enable-isis'. The terminal window is overlaid on a desktop environment with a taskbar at the bottom showing various application icons and a system clock indicating 9:43 AM on 4/14/2026.

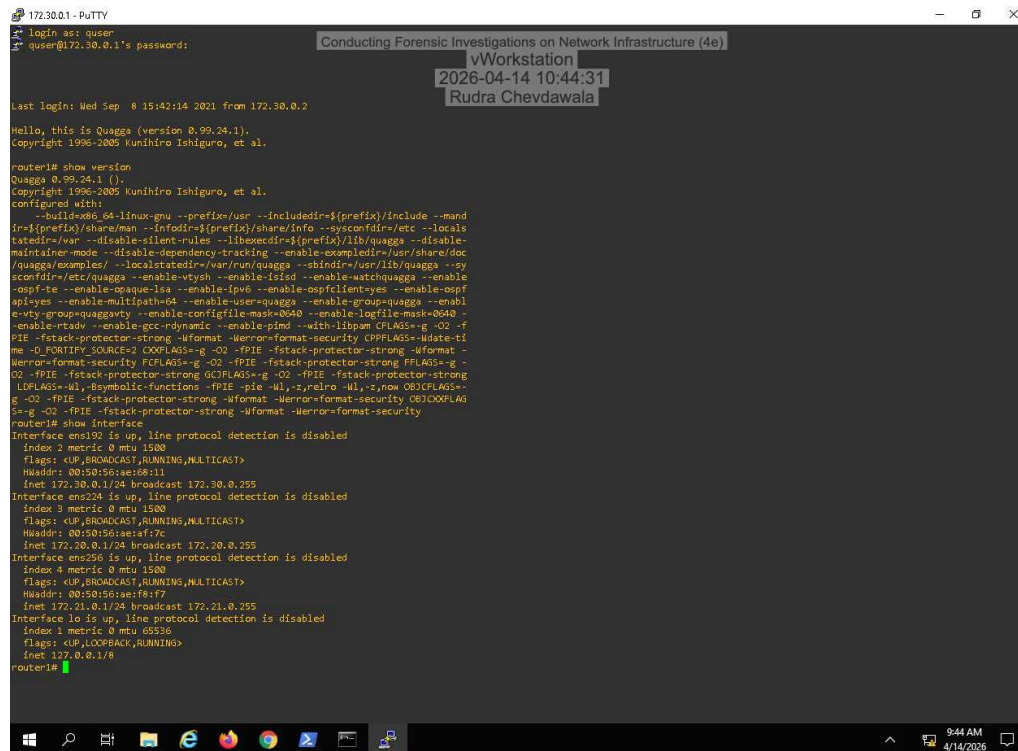
```
172.30.0.1 - PuTTY
login as: quser
quser@172.30.0.1's password:

Last login: Wed Sep  8 15:42:14 2021 from 172.30.0.2

Hello, this is Quagga (version 0.99.24.1).
Copyright 1996-2005 Kunihiro Ishiguro, et al.

router1a show version
Quagga 0.99.24.1 ().
Copyright 1996-2005 Kunihiro Ishiguro, et al.
configured with:
  --buildsys=64-linux-gnu --prefix=/usr --includedir=${prefix}/include --mandir=${prefix}/share/man --infodir=${prefix}/share/info --sysconfdir=/etc --localstatedir=/var --disable-silent-rules --libexecdir=${prefix}/lib/quagga --disable-maintainer-mode --disable-dependency-tracking --enable-examplesdir=/usr/share/doc/quagga/examples/ --localstatedir=/var/run/quagga --shindir=/usr/lib/quagga --sysconfdir=/etc/quagga --enable-vtysh --enable-isis --enable-watchquagga --enable-ospf-te --enable-ospf-lsa --enable-ipv6 --enable-ospfclienteyes --enable-ospf-apikeys --enable-multipath=64 --enable-user=quagga --enable-group=quagga --enable-vty-group=quagga --enable-configfile-mask=0640 --enable-logfile-mask=0640 --enable-rtadv --enable-gcc-rdynamic --enable-pimd --with-libpam CFLAGS=-g -O2 -fPIE -fstack-protector-strong -Wformat -Werror=format-security CPPFLAGS=-Wdate-time -D_FORTIFY_SOURCE=2 COXFLAGS=-g -O2 -fPIE -fstack-protector-strong -Wformat -Werror=format-security FFLAGS=-g -O2 -fPIE -fstack-protector-strong FFLAGS=-g -O2 -fPIE -fstack-protector-strong GCOFLAGS=-g -O2 -fPIE -fstack-protector-strong LDFLAGS=-Wl,-Bsymbolic-functions -fPIE -pie -Wl,-z,relro -Wl,-z,now O3CFLAGS=-g -O2 -fPIE -fstack-protector-strong -Wformat -Werror=format-security O3COXFLAGS=-g -O2 -fPIE -fstack-protector-strong -Wformat -Werror=format-security
router1a
```


7. Make a screen capture showing the router's interface details.



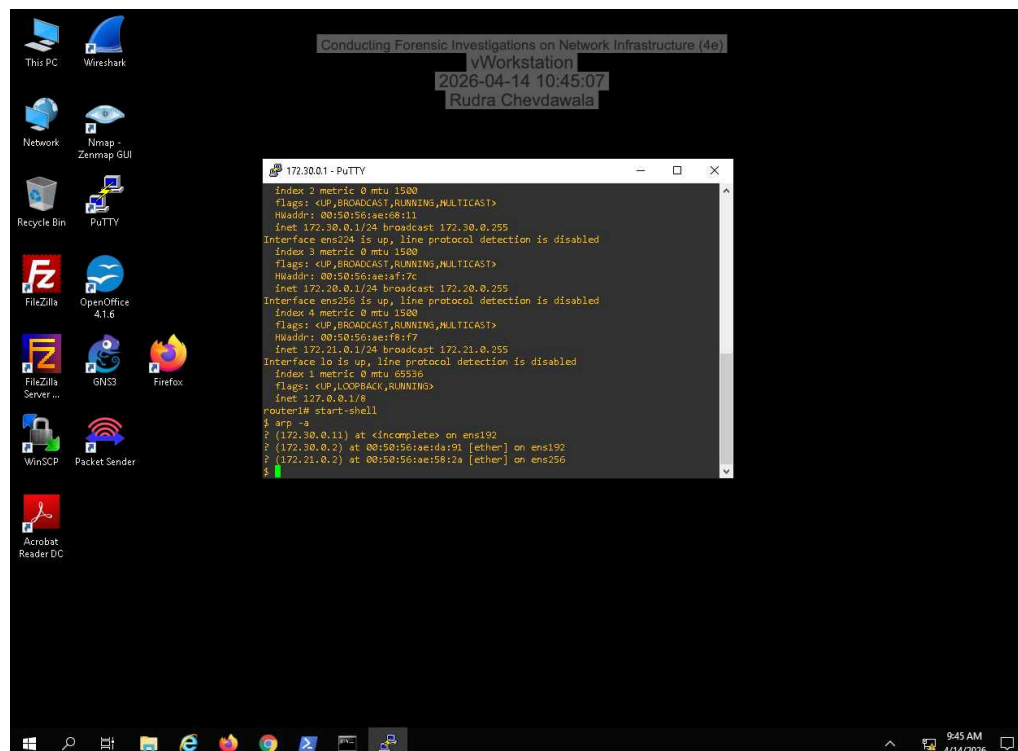
```
172.30.0.1 - PuTTY
login as: quser
quser@172.30.0.1's password:

Last login: Wed Sep  8 15:42:14 2021 from 172.30.0.2

Hello, this is Quagga (version 0.99.24.1).
Copyright 1996-2005 Kunihiro Ishiguro, et al.

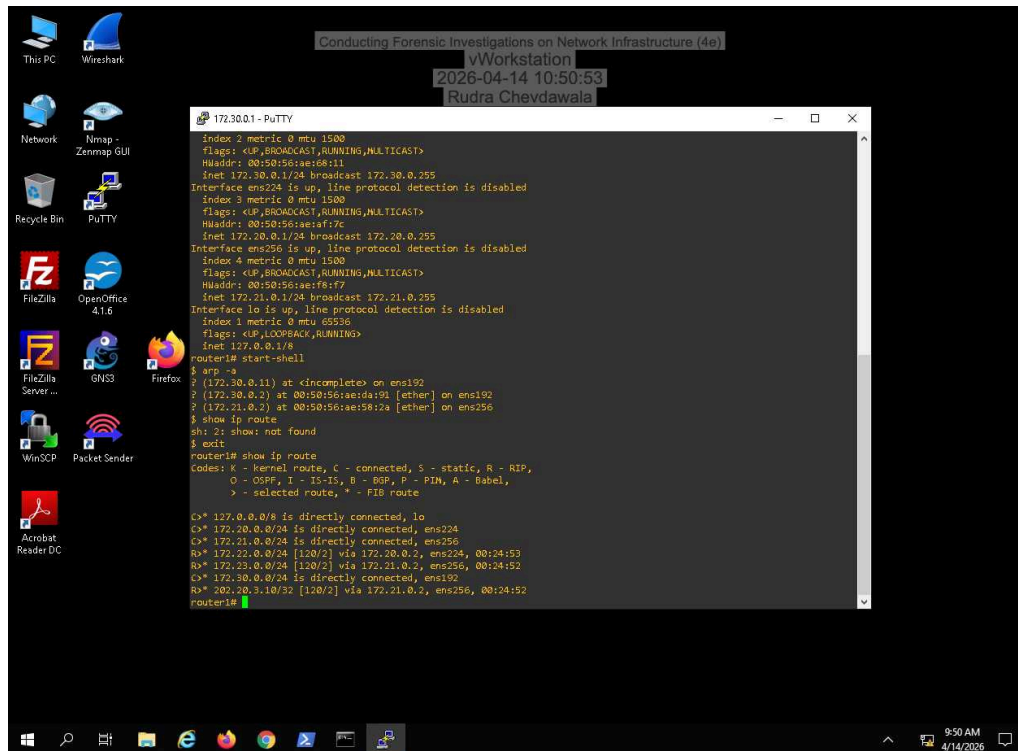
router1# show version
Quagga 0.99.24.1 ().
Copyright 1996-2005 Kunihiro Ishiguro, et al.
configured with:
  --buildsys=64-linux-gnu --prefix=/usr --includedir=${prefix}/include --mandir=${prefix}/share/man --infodir=${prefix}/share/info --sysconfdir=/etc --localstatedir=/var --disable-silent-rules --libexecdir=${prefix}/lib/quagga --disable-maintainer-mode --disable-dependency-tracking --enable-examplesdir=/usr/share/doc/quagga/examples --localstatedir=/var/run/quagga --shindir=/usr/lib/quagga --sysconfdir=/etc/quagga --enable-vtysh --enable-isis --enable-watchquagga --enable-ospf-te --enable-ospf-lsa --enable-ipv6 --enable-ospfclienteyes --enable-ospf-aplkeys --enable-multipath=64 --enable-user=quagga --enable-group=quagga --enable-vty-group=quagga --enable-configfile-mask=0640 --enable-logfile-mask=0640 --enable-rtadv --enable-gcc-rdynamic --enable-pimd --with-libpam CFLAGS=-g -O2 -fPIE -fstack-protector-strong -Wformat -Werror=format-security CPPFLAGS=-Wdate-time -D_FORTIFY_SOURCE=2 COXFLAGS=-g -O2 -fPIE -fstack-protector-strong -Wformat -Werror=format-security FFLAGS=-g -O2 -fPIE -fstack-protector-strong FFLAGS=-g -O2 -fPIE -fstack-protector-strong GCOFLAGS=-g -O2 -fPIE -fstack-protector-strong LDFLAGS=-Wl,-Bsymbolic-functions -fPIE -pie -Wl,-z,relro -Wl,-z,now O3 COXFLAGS=-g -O2 -fPIE -fstack-protector-strong -Wformat -Werror=format-security DB3COXFLAGS=-g -O2 -fPIE -fstack-protector-strong -Wformat -Werror=format-security DB3COXFLAGS=-g -O2 -fPIE -fstack-protector-strong -Wformat -Werror=format-security
router1# show interface
Interface ens192 is up, line protocol detection is disabled
  Index 2 metric 0 mtu 1500
  flags: <UP,BROADCAST,RUNNING,MULTICAST>
  Hwaddr: 00:50:56:a6:18:11
  Inet 172.30.0.1/24 broadcast 172.30.0.255
Interface ens224 is up, line protocol detection is disabled
  Index 3 metric 0 mtu 1500
  flags: <UP,BROADCAST,RUNNING,MULTICAST>
  Hwaddr: 00:50:56:a6:f7:c
  Inet 172.20.0.1/24 broadcast 172.20.0.255
Interface ens256 is up, line protocol detection is disabled
  Index 4 metric 0 mtu 1500
  flags: <UP,BROADCAST,RUNNING,MULTICAST>
  Hwaddr: 00:50:56:a6:f8:f7
  Inet 172.21.0.1/24 broadcast 172.21.0.255
Interface lo is up, line protocol detection is disabled
  Index 1 metric 0 mtu 65536
  flags: <UP,LOOPBACK,RUNNING>
  Inet 127.0.0.1/8
router1#
```

10. Make a screen capture showing the router1 ARP table.

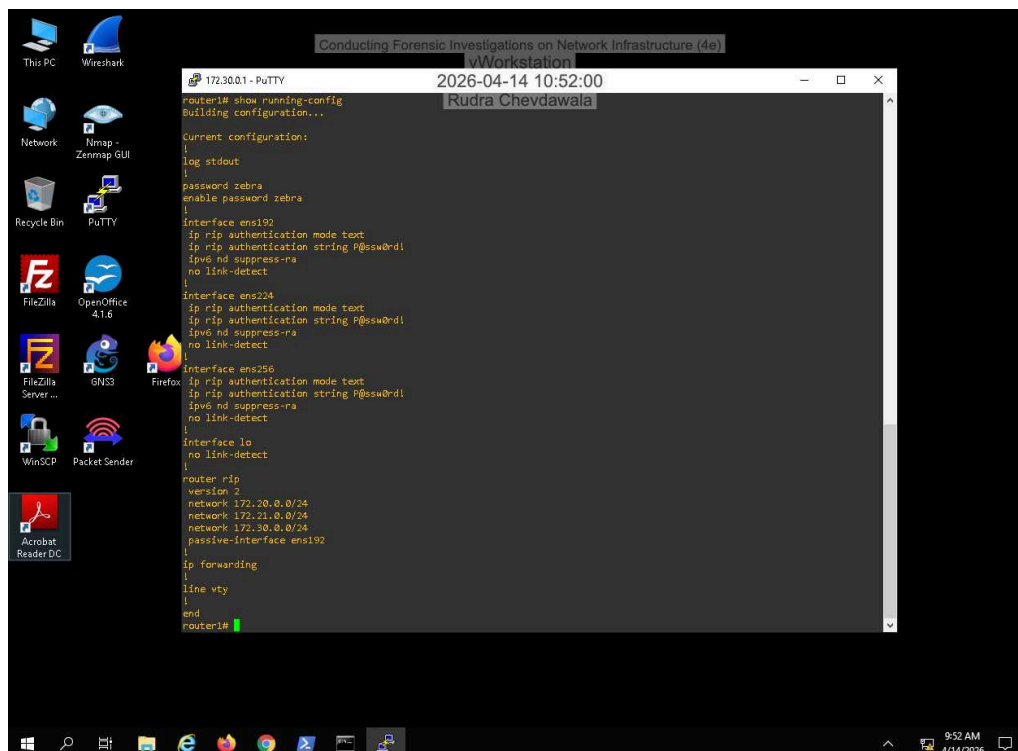


```
172.30.0.1 - PuTTY
Index 2 metric 0 mtu 1500
flags: <UP,BROADCAST,RUNNING,MULTICAST>
Hwaddr: 00:50:56:a6:18:11
Inet 172.30.0.1/24 broadcast 172.30.0.255
Interface ens224 is up, line protocol detection is disabled
Index 3 metric 0 mtu 1500
flags: <UP,BROADCAST,RUNNING,MULTICAST>
Hwaddr: 00:50:56:a6:f7:c
Inet 172.20.0.1/24 broadcast 172.20.0.255
Interface ens256 is up, line protocol detection is disabled
Index 4 metric 0 mtu 1500
flags: <UP,BROADCAST,RUNNING,MULTICAST>
Hwaddr: 00:50:56:a6:f8:f7
Inet 172.21.0.1/24 broadcast 172.21.0.255
Interface lo is up, line protocol detection is disabled
Index 1 metric 0 mtu 65536
flags: <UP,LOOPBACK,RUNNING>
Inet 127.0.0.1/8
router1# start-shell
$ arp -a
? (172.30.0.11) at <incomplete> on ens192
? (172.30.0.2) at 00:50:56:a6:18:11 [ether] on ens192
? (172.21.0.2) at 00:50:56:a6:f8:f7 [ether] on ens256
$
```

13. Make a screen capture showing the IP routing table.



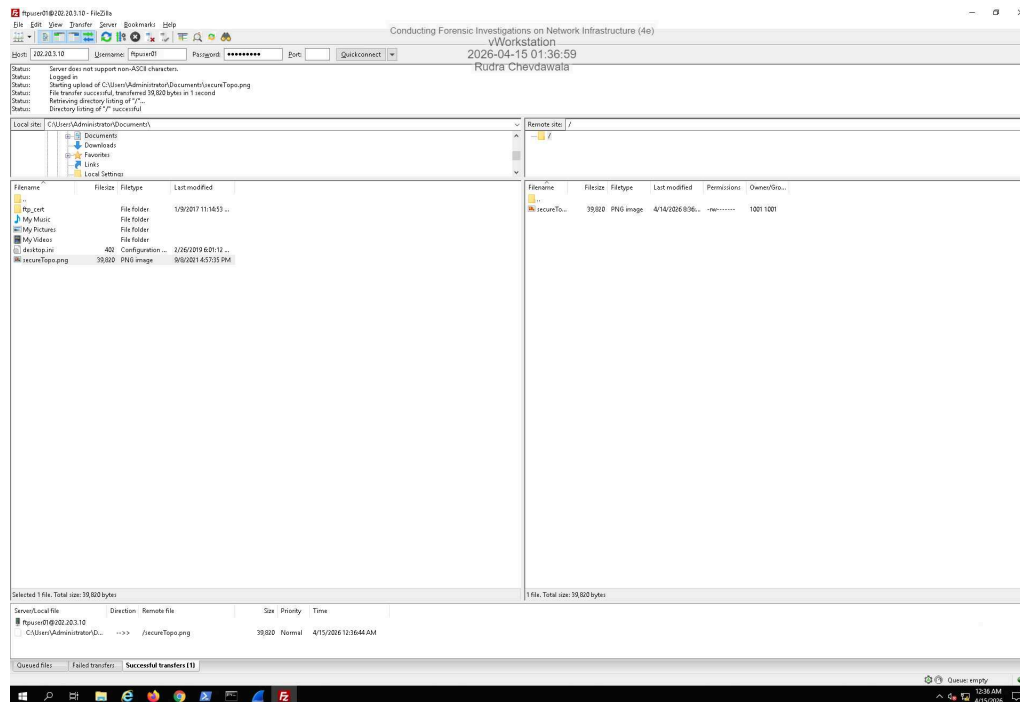
15. Make a screen capture showing the currently running configuration.



Section 2: Applied Learning

Part 1: Perform Advanced Packet Capture and Analysis

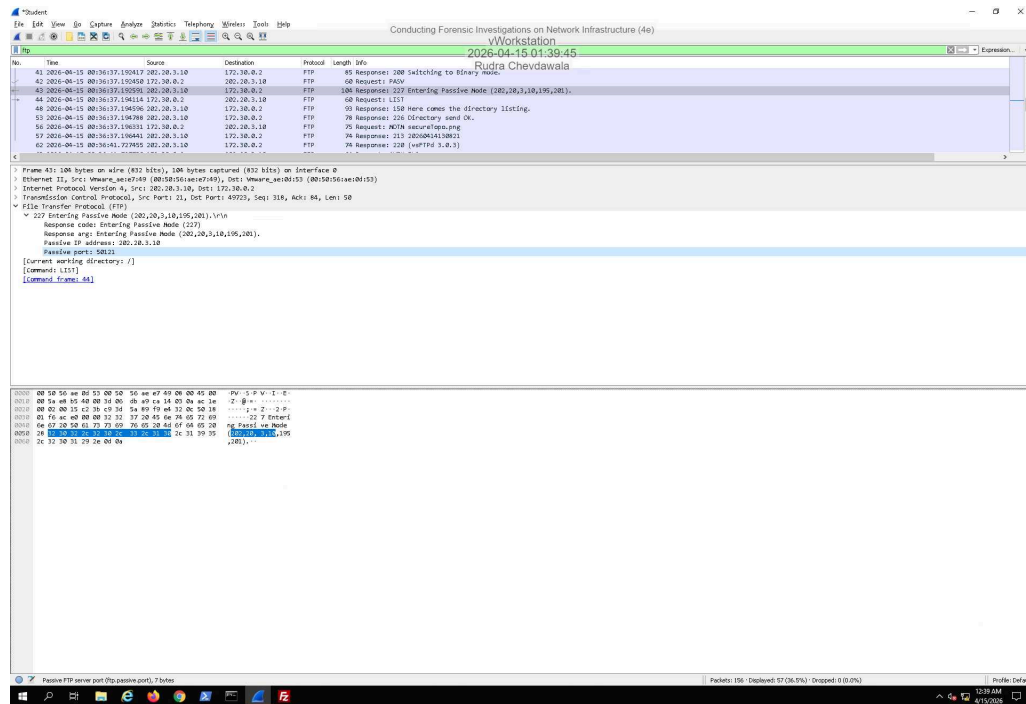
7. Make a screen capture showing the **successful transfer of the secureTopo.png file**.



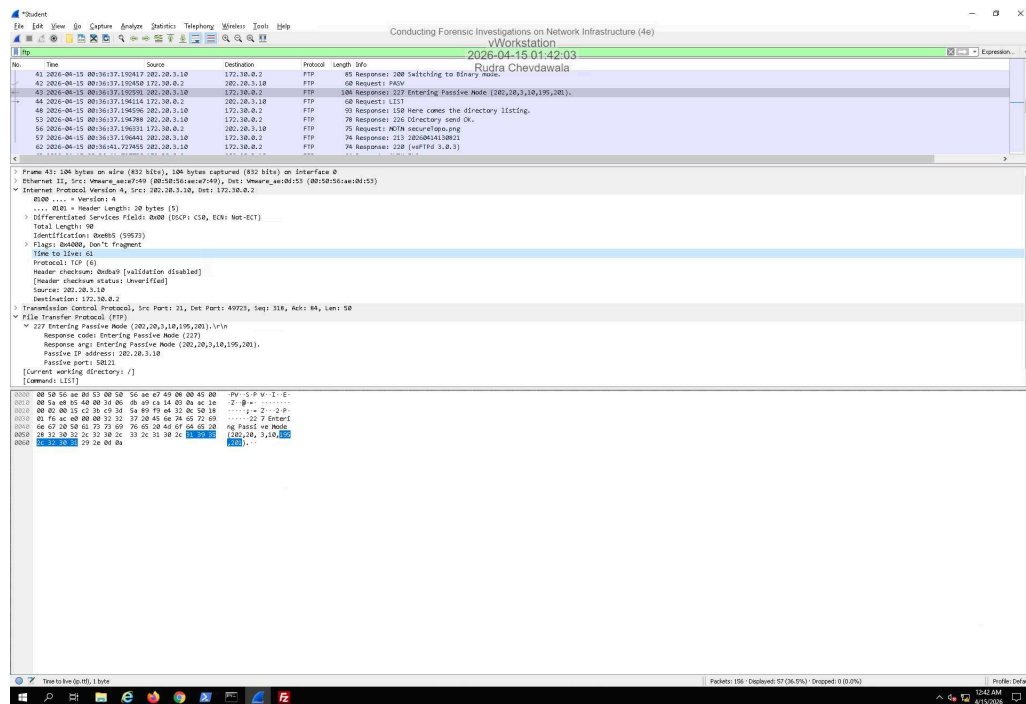
Conducting Forensic Investigations on Network Infrastructure (4e)

Digital Forensics, Investigation, and Response, Fourth Edition - Lab 09

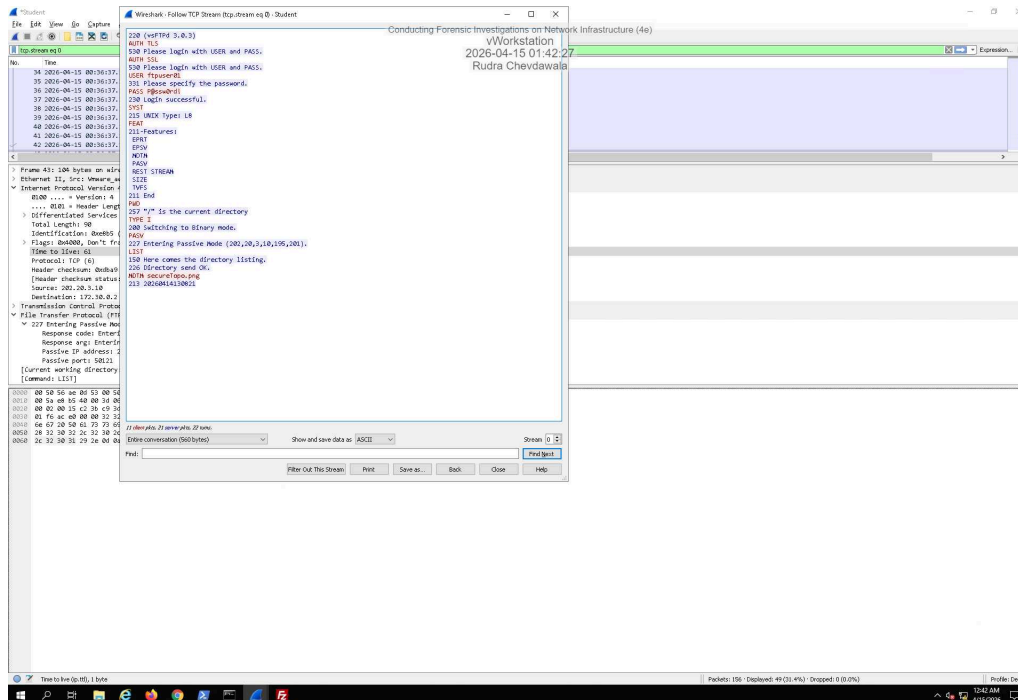
15. Make a screen capture showing the passive port specified by the FTP server in the Packet Details pane.



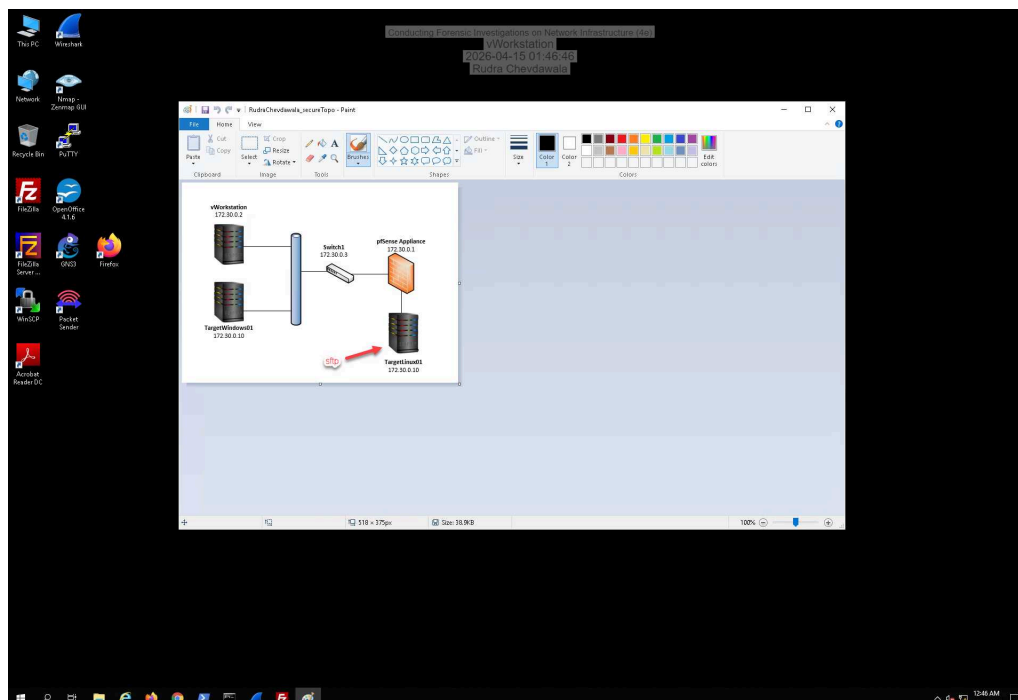
18. Make a screen capture showing the Time to live field in the Packet Details pane.



20. Make a screen capture showing the Follow TCP stream window.

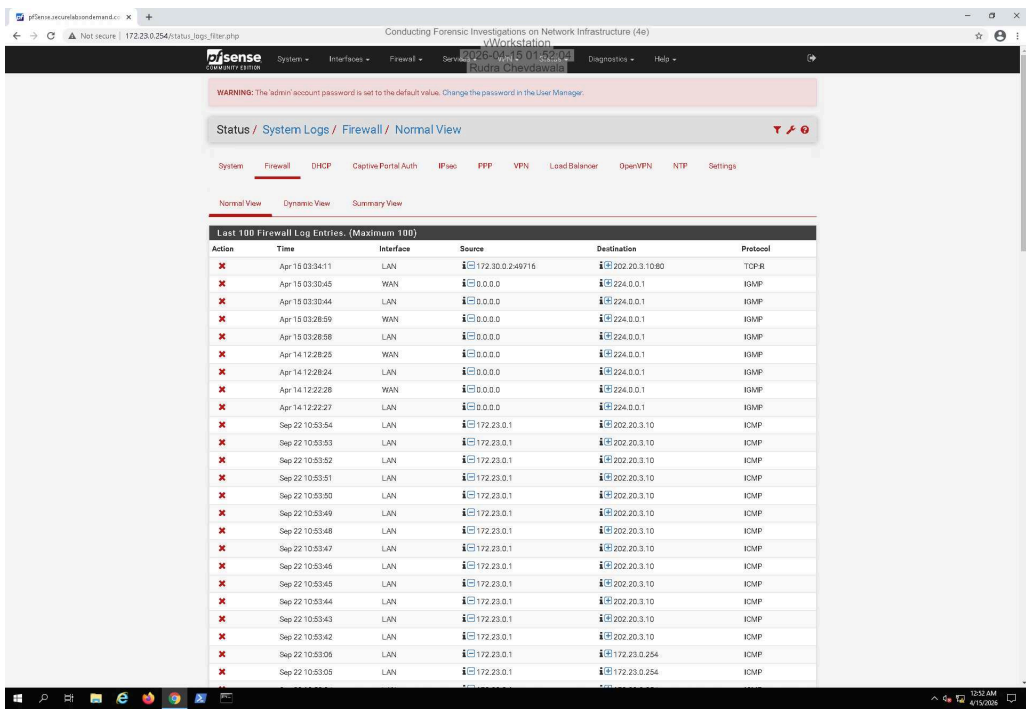


32. Make a screen capture showing the reconstituted PNG file.

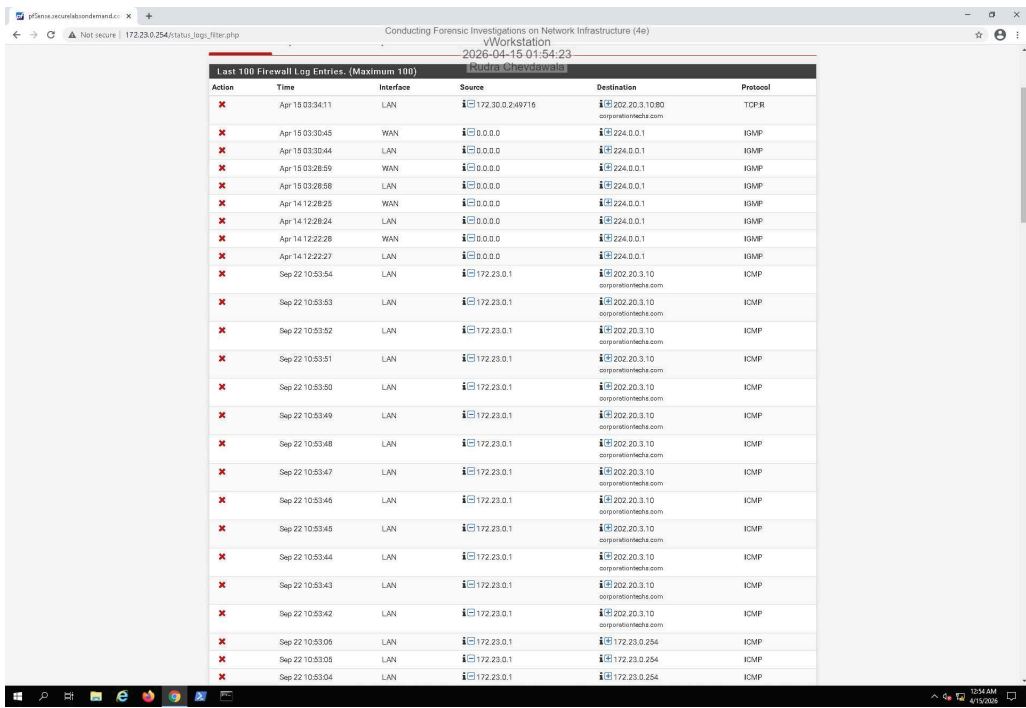


Part 2: Analyze a Firewall for Forensic Evidence

9. Make a screen capture showing the entries in the firewall log.



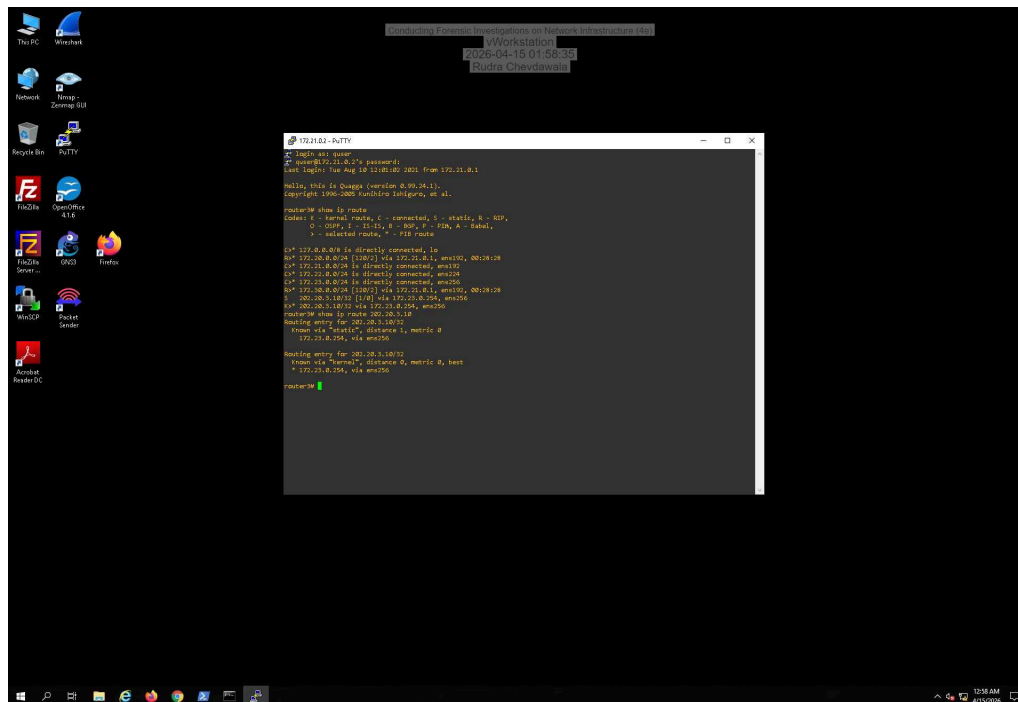
11. Make a screen capture showing the resolved entries in the firewall log.



Section 3: Challenge and Analysis

Part 1: Identify the Source of a Suspicious Route

Make a screen capture showing the non-RIP route that you discovered on the target router.



Part 2: Identify Suspicious Outgoing Connections

Record the destination IP address and Port number of the outgoing connection attempt.

The destination IP address is 202.20.3.10, and the outgoing connection attempt used port 1337.